

B. Tech. Semester I/II

Subject Name: Basic Electrical Engineering

Subject Code: BTTEL12108

Type of course: Engineering Science

Prerequisite: NA.

Rationale: Electricity has been the main source of energy for the developing and developed countries. Per capita consumption of electricity of a country can be considered as an indicator of the development of the country. In view of this, it is essential for all engineering graduates to know the basic aspects of electrical engineering. This subject deals with basic circuit solution methods, single-phase and three-phase AC circuits, magnetic circuits and transformers, introduction to electrical machines, introduction to batteries, illumination schemes and Electrical safety.

Teaching and Examination Scheme:

TEACHING SCHEME				Theory Marks			Practical Marks		Total
L	T	P	C	TEE	CA1	CA2	TEP	CA3	
3	0	2	4	60	25	15	30	20	150

CA1: Continuous Assessment (assignments/projects/open book tests/closed book tests **CA2:** Sincerity in attending classes/class tests/ timely submissions of assignments/self-learning attitude/solving advanced problems **TEE:** Term End Examination **TEP:** Term End Practical Exam (Performance and viva on practical skills learned in course) **CA3:** Regular submission of Lab work/Quality of work submitted/Active participation in lab sessions/viva on practical skills learned in course

Content:

Sr. No.	Topics	Teaching Hrs.	Module Weightage
1.	DC Circuits: Electrical circuit elements (R, L and C), voltage and current sources, Kirchoff's current and voltage laws, analysis of simple circuits with dc excitation using Loop Current and Nodal Voltage techniques, Source Conversion, Star-delta conversion, Superposition theorem, Thevenin's and Norton's theorems, Series-parallel combination of capacitors, energy stored in capacitor, Time-domain analysis of first-order RL and RC circuits.	10	20%
2.	Single-phase and Three-Phase AC Circuits	12	25%

Sr. No.	Topics	Teaching Hrs.	Module Weightage
	Representation of sinusoidal waveforms, peak and RMS values, phasor representation of AC quantities, real power, reactive power, apparent power, power factor. Analysis of single-phase ac circuits consisting of R, L and C, series-parallel combinations of R-L-C circuit, series and parallel resonance. Three-Phase balanced circuits, voltage and current relations in star and delta connections, Power measurement in three-phase circuits, Basics of Power factor improvement.		
3.	Magnetic Circuit and Transformer Magnetic materials, B-H characteristics, Series and Parallel magnetic circuit, Concepts of self-inductance, mutual inductance and coefficient of coupling; Inductance in series and parallel; Statically and dynamically induced EMF; Hysteresis and Eddy current losses; Energy stored in magnetic fields. Construction and working principle of single-phase and three-phase transformers, Ideal and practical transformer, efficiency and voltage regulation of transformer, Auto-transformer and its applications.	8	20%
4.	Electrical Machines Generation of rotating magnetic fields; Construction, working principle, types, operating characteristics and applications of following machines: • Single-Phase and Three-Phase Induction motor • DC Motor • Synchronous generator	8	20%
5.	Electrical Installations and Safety Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of wires and cables, Types of Batteries, Important characteristics for batteries, earthing, electrical safety precautions, electric shock and first-aid for electric shock, other hazards of electrical laboratories & safety rules.	7	15%

Sarvajani University
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Suggested Specification Table of Marks as per Bloom's Taxonomy (Theory/Practical):

% Distribution of Marks					
R Level	U Level	A Level	N Level	E Level	C Level
40	20	20	20	-	-

Legends: R: Remembrance, U: Understanding; A: Application, N: Analyze, E: Evaluate C: Create and above Levels

Note: This specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Reference Text Books:

Sr. No.	Title of book /article	Author(s)	Publisher and details like ISBN	Year of publication	Publication Edition
1.	Basic Electrical Engineering	D. C. Kulshreshtha,	McGraw Hill,	2009	First
2.	Electrical Technology: Vol1 Basic Electrical Engineering	B. L. Theraja	S. Chand and Co.	2005	First Multicolour
3.	Electrical Technology: Vol2 AC & DC Machines	B. L. Theraja	S. Chand and Co.	2005	First Multicolour
4.	Basic Electrical Engineering	I. J. Nagrath and D. P. Kothari	Tata McGraw Hill	2010	Second
5.	Electrical Machines	D. P. Kothari and I. J. Nagrath	Tata McGraw Hill	2010	
6.	Electrical Engineering Fundamentals	V. D. Toro	Prentice Hall India	2016	Second
7.	Basic Electrical Engineering	V. N. Mittle	McGraw Hill	1995	

Course Outcome:

Sr. No.	CO Statement After learning this subject, students will be able to	Marks % weightage
CO-1	Apply fundamental electrical laws and circuit theorems in solving electrical circuits. (<i>R, U - Cognitive level</i>)	20
CO-2	Analyze single-phase and three-phase AC circuits. (<i>U, A - Cognitive level</i>)	25
CO-3	Comprehend magnetic circuit, principle and application of transformer. (<i>R, U, A, N - Cognitive level</i>)	20
CO-4	Describe operating principle and applications of electrical machines. (<i>U, A - Cognitive level</i>)	20
CO-5	Comprehend electrical installation components, types and characteristics of batteries, importance of safety and the precaution while working with electrical equipment. (<i>R, U, A, N - Cognitive level</i>)	15

Mapping with POs:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO-1	3	2	-	3	2	-	-	-	-	-	-	3			
CO-2	3	2	2	3	2	-	2	-	-	-	-	3			
CO-3	3	2	-	3	2	-	-	-	-	-	-	3			
CO-4	3	2	-	3	2	-	-	-	-	-	-	3			
CO-5	3	2	3	3	2	3	-	3	-	-	-	3			
Rationale*	15	10	6	15	10	3	2	3	-	-	-	15			

***Rationale:** Explaining why it is matching this particular program outcome

LIST OF PRACTICALS: (Minimum ---10--- performed.)

- (1) Introduction and use of measuring instruments – voltmeter, ammeter, multi-meter, oscilloscope. Resistors, Capacitors and Inductors, basic safety precautions.
- (2) To verify Kirchoff's current and voltage laws.
- (3) To verify superposition theorem for DC circuit.
- (4) To verify Thevenin's theorem.
- (5) To verify Norton's theorem.
- (6) To obtain sinusoidal steady state response of R-L and R-C circuits – impedance calculation

- and verification. Observation of phase differences between current and voltage.
- (7) To measure power in single-phase R-L and R-C series circuits.
 - (8) To verify the resonance in R-L-C circuits.
 - (9) To measure the power in three-phase circuits using two wattmeter method.
 - (10) To verify the current and voltage relationships in three phase star and delta connections.
 - (11) To perform o. c./s. c. test of transformer.
 - (12) Demonstration of cut-section models and charts of various machines.
 - (13) To verify the power factor improvement in single phase AC circuit.

Major Equipment:

Ammeters, Voltmeters, Wattmeter's, Resistors, Capacitors and Inductors of appropriate rating, Multi-meters, Digital storage oscilloscope, Power supply, Cut section models/charts of various machines, Charts for earthing and safety precaution.

List of Open Source/learning website:

1. MIT OPEN COURSEWARE by Massachusetts Institute of Technology - website: ocw.mit.edu
2. Courses available through NPTEL. - Website: nptel.ac.in

List of Open Source Software:

Website: www.vlabs.co.in